



Access S3 from a VPC

GE

geoayodele2011@gmail.com

```
aws [Search] [Alt+S] Ask Amazon Q
ec2-user@ip-10-0-6-108 ~]$ aws s3 ls
unable to locate credentials. You can configure credentials by running "aws login".
ec2-user@ip-10-0-6-108 ~]$ aws configure
AWS Access Key ID [None]: AKIAW2F22MTETJF3UGUR
AWS Secret Access Key [None]: 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
default region name [None]: us-east-1
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default region name [us-east-1]: us-east-1
default output format [us-east-1]: [ec2-user@ip-10-0-6-108 ~]$ aws s3 ls
2026-05-04 17:18:18 nextwork-vpc-project-olowolafe
ec2-user@ip-10-0-6-108 ~]$ ^C
ec2-user@ip-10-0-6-108 ~]$ aws s3 ls s3://nextwork-vpc-project-olowolafe
2026-05-04 17:22:38 156481 Screenshot (260).png
2026-05-04 17:20:30 13823 Theory TYPING.docx
ec2-user@ip-10-0-6-108 ~]$ ^C
ec2-user@ip-10-0-6-108 ~]$ sudo touch /tmp/test.txt
ec2-user@ip-10-0-6-108 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-olowolafe
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-olowolafe/test.txt
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2026-05-04 18:13:24 0 test.txt
ec2-user@ip-10-0-6-108 ~]$
```



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a logically isolated, private network within a public cloud provider's infrastructure. It is used to securely run cloud resources—such as virtual machines, databases, and containers.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC (Virtual Private Cloud) as the foundational network environment for my EC2 instance and S3 interactions.

One thing I didn't expect in this project was...

One thing I probably didn't expect in this project was how credentials became the main challenge when connecting your EC2 instance to S3.

This project took me...

This project took me about three hours.



In the first part of my project...

Step 1 - Architecture set up

In this step, I will demonstrate how to build a Virtual Private Cloud (VPC) from scratch and launch an EC2 instance into it. I am doing this project to learn how cloud networking works in practice, strengthen my understanding of AWS services, and gain hands-on experience with configuring secure, scalable infrastructure. By starting with the fundamentals, I will develop the skills needed to design and manage cloud environments that can support real-world applications.because...

Step 2 - Connect to my EC2 instance

In this step, I will connect directly to my EC2 instance using SSH, because this allows me to verify that the virtual server I launched inside my custom VPC is reachable and functioning properly. By logging in, I can begin interacting with the instance, test network connectivity, and prepare it for deploying applications or accessing other AWS services.

Step 3 - Set up access keys

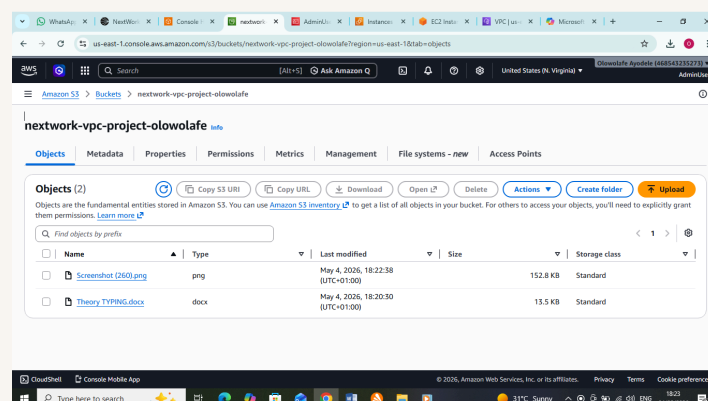
In this step, I will create an AWS access key, because it provides the authentication credentials (Access Key ID and Secret Access Key) that allow us to securely interact with AWS services using the CLI or SDK



Architecture set up

I started my project by launching an Amazon EC2 instance into my newly created VPC. This gave me a virtual server running inside the isolated network environment I designed. Along with the instance, I also set up supporting resources such as a subnet, an internet gateway, a route table, and a security group. Together, these resources ensure that my EC2 instance is properly connected, secure, and ready for me to test connectivity and deploy applications.

In this step i also set up a bucket in Amazon S3 and uploaded two files, because this establishes a secure storage resource that I can later access from my EC2 instance to practice integrating compute with storage





AWS CLI is The AWS Command Line Interface (AWS CLI) is a unified tool that lets you manage AWS services directly from your terminal, instead of using the web console. It's especially useful for automation, scripting, and working with AWS resources at scale I have access to AWS CLI because because this allows me to verify that the virtual server I launched inside my custom VPC is reachable, secure, and functioning properly

The first command I ran was `aws s3 ls`. This command is used to list all of the S3 buckets in your account.

The second command I ran was `aws configure`. This command is used to set up your AWS CLI with the credentials and default settings it needs to talk to AWS services.

[illegible]



Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, essentially connecting compute to the rest of my cloud resources.

Access keys in AWS are a pair of security credentials that allow you to authenticate programmatically when using tools like the AWS CLI, SDKs, or APIs.

Secret access keys are your private credentials that, paired with the Access Key ID, allow secure programmatic access to AWS services.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use IAM roles with temporary credentials instead of long-term static keys.



In the second part of my project...

Step 4 - Set up an S3 bucket

In this step, I will In this step, I will create a bucket in Amazon S3, because it provides a secure and scalable place to store objects that I can later access and manage directly from my EC2 instance.”

Step 5 - Connecting to my S3 bucket

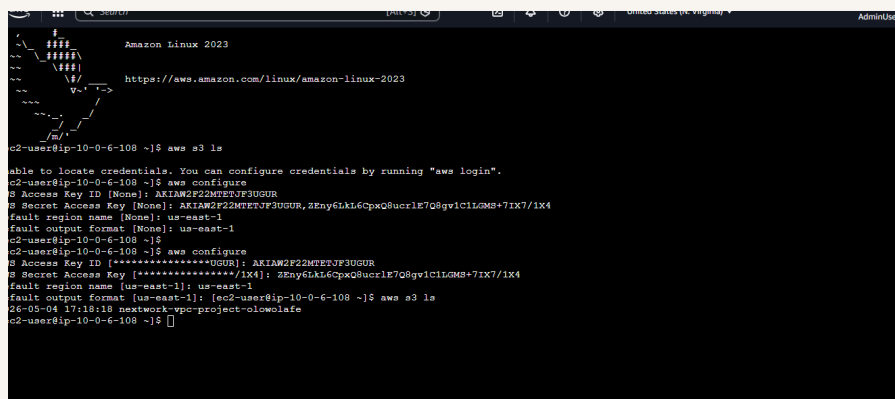
In this step, I will head back to my EC2 instance and get it to interact with my S3 bucket, because this shows how AWS compute resources can securely access and manage storage resources.



Connecting to my S3 bucket

The first command I ran was `aws s3 ls`. This command is used to list all of the S3 buckets in your account.

When I ran the command `aws s3 ls`, again, the terminal responded with `nextwork-vpc-project-olowolafe`. This indicated that my VPC and other resources are connected.



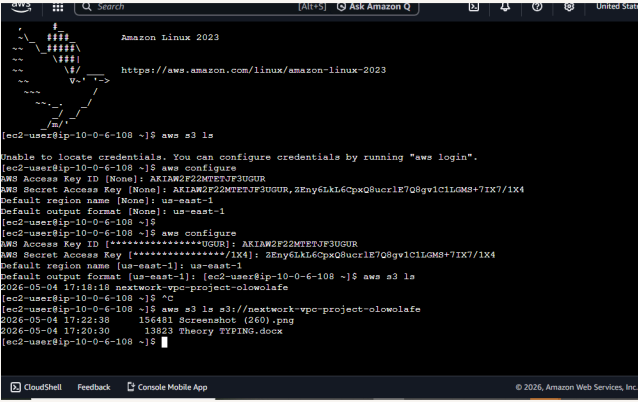
```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

c2-user@ip-10-0-6-108 ~]$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws login".
c2-user@ip-10-0-6-108 ~]$ aws configure
S Access Key ID [None]: AKIAW2F22MTETJF3UGUR
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fault region name [None]: us-east-1
fault output format [None]: us-east-1
c2-user@ip-10-0-6-108 ~]$ aws configure
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26-05-04 17:18:18 nextwork-vpc-project-olowolafe
c2-user@ip-10-0-6-108 ~]$
```




Connecting to my S3 bucket

Another CLI command I ran was 'aws s3 ls s3://nextwork-vpc-project-olowolafe' which returned the two files I uploaded earlier.



```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-6-108 ~]$ aws s3 ls
Unable to locate credentials. You can configure credentials by running "aws login".
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[ec2-user@ip-10-0-6-108 ~]$
```



Uploading objects to S3

To upload a new file to my bucket, I first ran the command `sudo touch /tmp/test.txt`. This command creates will create an empty file.

The second command I ran was `aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-olowolafe`. This command will copy files into my bucket.

The third command I ran was `aws s3 ls s3://nextwork-vpc-project-olowolafe` which validated that the object in my bucket

```
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Unable to locate credentials. You can configure credentials by running "aws login".
sc2-user@ip-10-0-6-108 ~$ aws configure
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```



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